



University of Salford, MSc Data Science

Module: Machine Learning & Data Mining

Session: Workshop Week 9

Topic: Ensemble Learning using AutoML

Tools: Azure Machine Learning Studio

Objectives:

After completing this workshop, you will be able to:

- Upload and use a local file in AzureML
- Use AutoML for deploying classification models
- Apply Ensemble models on datasets
- Assess and evaluate model performance

Ensemble Learning

In ensemble learning you combine multiple learning algorithms to obtain better predictive performance than could be obtained from any of the learning algorithms alone. This is like when a patient refers to multiple specialist doctors to diagnosis a disease rather than relies on one doctor. In this workshop, you will learn how to use AzureML for ensemble learning. AzureML helps you to build machine learning models using the designer without writing a single line of code.

Create an Azure Machine Learning workspace

1. Sign into the [Azure portal](https://portal.azure.com/) using your Microsoft credentials. <https://portal.azure.com/>
2. Select + **Create a resource**, search for *Machine Learning*, and **create** a new **Azure Machine Learning** resource with an *Azure Machine Learning* plan. Use the following settings:
 - **Subscription:** Your Azure subscription (Azure for Students)
 - **Resource group:** Click on create new and give it a name.
 - **Workspace name:** Enter a unique name for your workspace.
 - **Region:** Select the closest geographical region (Select Italy north)
 - **Storage account:** Note the default new storage account that will be created for your workspace (Don't change it)
 - **Key vault:** Note the default new key vault that will be created for your workspace (Don't change it)
 - **Application insights:** Note the default new application insights resource that will be created for your workspace (Don't change it).
 - **Container registry:** None (one will be created automatically the first time you deploy a model to a container).
3. Select **Review + create**, then click on **Create**. Wait for your workspace to be created (it can take a few minutes) and then go to the deployed resource by clicking on **Go to resource**.
4. Scroll down and select **Launch studio** (or open a new browser tab and navigate to <https://ml.azure.com>, and sign into Azure Machine Learning studio using your Microsoft account).
5. If the “What are your machine learning goals today?” message appears, select **Cancel**.
6. If the **Welcome to the studio!** message appears, select **X**.
7. In Azure Machine Learning studio, you should see your newly created workspace. If that is not the case, click on **Microsoft** on the left-hand menu. Then from the new

left-hand menu, select **All Workspaces**, where all the workspaces associated with your subscription are listed. Choose the one you created for this exercise.

Your Azure subscription will be charged a small amount for data storage as long as the Azure Machine Learning workspace exists in your subscription, **so we recommend you delete the Azure Machine Learning workspace when it is no longer required at the end of workshop.**

Create compute

1. In [Azure Machine Learning studio](#), select the three lines at the top left to view the various pages in the interface (you may need to maximize the size of your screen). You can use these pages in the left-hand pane to manage the resources in your workspace. Select the **Compute** page (under **Manage**).
2. On the **Compute** page, select the **Compute clusters** tab, and click on '+New' button to add a new compute cluster with the following settings. You'll use this to train a machine learning model:
 - Virtual machine type: CPU
 - Virtual machine size:
 - Choose Select from all options
 - Search for and select **Standard_DS11_v2**
 - Select **Next**
 - **Compute name**: enter a unique name.
 - Set "Minimum number of nodes" as 0 and "Maximum number of nodes" as 2 and "Idle seconds before scale down" as 120
 - Enable SSH access: Clear (off)
 - Select **Create**

The compute cluster will take some time to be created. You can move onto the next step while you wait.

Note Compute instances and clusters are based on standard Azure virtual machine images. For this module, the *Standard_DS11_v2* image is recommended to achieve the optimal balance of cost and performance. If your subscription has a quota that does not include this image, choose an alternative image; but bear in mind that a larger image may incur higher cost and a smaller image may not be sufficient to complete the tasks. Alternatively, ask your Azure administrator to extend your quota.

Create a dataset

In this workshop, we will use the Diabetes data we had in the previous session. But in this week, you will learn how to upload data to Azure from your local computer instead of using web URL. So, download the diabetes.csv file from the blackboard and then follow the steps:

1. In [Azure Machine Learning studio](#), expand the left pane by selecting the three lines at the top left of the screen. View the **Data** page (under **Assets**). The Data page contains specific data files or tables that you plan to work with in Azure ML. You can create datasets from this page as well.
2. Click on +Create
 - Enter basic info:
 - **Name:** Diabetes
 - **Description:** Local diabetes data
 - **Dataset type:** Tabular
 - Click **Next** and choose '**From local file**' and click **Next** again.
 - Choose workspaceblobstore and click next.
 - Click the upload files or folders button and choose upload files. Then select your csv file and click Next

Choose a file or folder

Choose files or folders to upload from your local drive. If you upload multiple folders or files, they will be stored in a containing folder.

Upload path

azureml://subscriptions/9ca1ddf6-6cd6-4531-9618-6a97d1049abf/resourcegroups/Week9/workspace... 

Upload 

☐ Overwrite if already exists

Upload list

diabetes.csv	 505.62 KB/505.62 KB	...
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- Settings and preview:
 - **File format:** Delimited
 - **Delimiter:** Comma
 - **Encoding:** UTF-8
 - **Column headers:** Only first file has headers
 - **Skip rows:** None
 - **Dataset contains multi-line data:** *do not select*
- Click next and then in the Schema:

- Include all columns other than **Path**
- Review the automatically detected types
- Click next and click create

After the dataset has been created, open it and view the **Explore** tab to see a sample of the data.

Diabetes Version: 1 (latest) ☆

Details Consume **Explore** Models Jobs

🔔 Profile: This is the quick profile generated by the top 10,000 rows. Please generate profile to view the schema and summary statistics for full data. ✕

🔄 Refresh ▶ Generate profile

Preview **Profile**

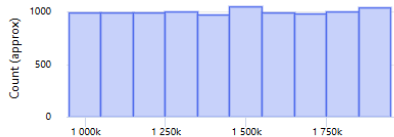
Number of columns: 10 Number of rows: First 50

PatientID	Pregnancies	PlasmaGlu...	DiastolicBl...	TricepsThi...	SerumInsu...	BMI	DiabetesP...	Age	Diabetic
1354778	0	171	80	34	23	43.51	1.213	21	0
1147438	8	92	93	47	36	21.241	0.158	23	0
1640031	7	115	47	52	35	41.512	0.079	23	0
1883350	9	103	78	25	304	29.582	1.283	43	1
1424119	1	85	59	27	35	42.605	0.55	22	0

In this window, you can click on Profile tab to see different information about each column.

Preview **Profile**

Number of columns: 10 Number of rows: 10000

Column	Profile	Type	Min	Max	Count	Missing count
PatientID		Integer	1000038	1999997	10000	0
Pregnancies		Integer	0	14	10000	0

Run an automated machine learning job

Exploring all possible combinations and coming up with a good machine learning model is tedious and time-consuming. **Azure AutoML** is a cloud-based service that can be used to automate machine learning pipelines building for classification, regression and forecasting tasks. Its goal is not only to tune hyper-parameters of a given model, but also to identify which model to use and how to pre-process the input dataset.

Follow the next steps to run a job that uses automated machine learning to train a classification model that predicts whether or not a person is diabetic.

1. In [Azure Machine Learning studio](#), select the **Automated ML** page (under **Authoring**).
2. Select the **+New Automated ML job** on the top left of the Automated ML view.
3. Create an Automated ML run with the following settings:
 - Change the Job name as predictdiabet and click Next
 - From “Select task type” select classification (it is the type of task that you would like your model to perform)
 - In Select dataset section, select Diabetes and then click Next

Select task type * ⓘ

Classification

Select dataset

Make sure your data is preprocessed into a supported format.

+ Create Refresh Show supported data assets only

Search Filter

Name	Type	Created on ↓	Modified on
✓ Diabetes	Table	Nov 11, 2023 11:38 AM	Nov 11, 2023 11:38 AM

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Back Next Cancel

- In next page, choose Diabetic as Target column (it shows which column is your class label)
- In “Classification settings” section, there are ‘View additional configuration settings’ and ‘View featurization settings’. Configure these settings by clicking on them.

Task type

Classification

Data

Diabetes ([View data](#))

Target column *

Diabetic (Integer) *

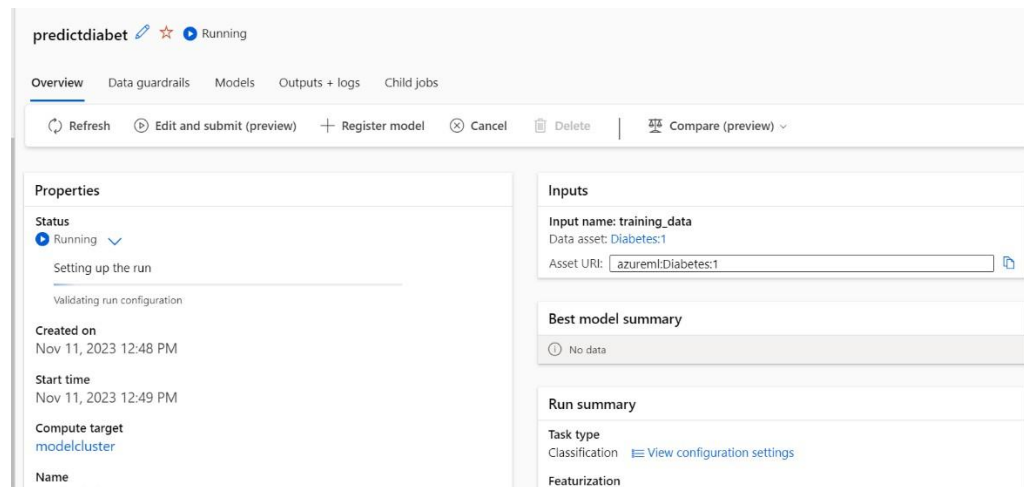
Classification settings

☐ Enable deep learning ⓘ

 View additional configuration settings  View featurization settings

- Click on View Additional configuration settings:
 - **Primary metric:** Select Accuracy.
 - **Explain best model:** Selected (this option causes automated machine learning to calculate feature importance for the best model which makes it possible to determine the influence of each feature on the predicted label.
 - **Enable ensemble stacking:** Unselected.
 - **Use all supported models:** Unselected (You'll restrict the job to try only a few specific algorithms)
 - **Allowed models:** Select only **KNN**, **RandomForest**, and **LightGBM** (you'd want to try as many as possible, but be aware that adding each model extends the duration required to execute the task)
 - Click on **Save**.
- Click on “View featurization settings” and configure it as follows:
 - **Enable featurization:** Selected (Automatically preprocess the features before training)
 - Click on **Save**.
- Select the validation and test type
 - Validation type: Automatic
 - Test dataset: train-test split for test dataset (Percentage test of data=20)
- Click Next.
- In “Select compute type” select compute cluster.
- You can choose an existing cluster or create a new one with the following configuration:
 - Location: Italy north
 - Virtual machine tier: Dedicated
 - Virtual machine type: CPU
 - Virtual machine size: Choose “Select from all options” and then select Standard_DS11_v2
 - Click Next.
 - Give a name to your Compute name
 - Set “Minimum number of nodes” as 0 and “Maximum number of nodes” as 2 and “Idle seconds before scale down” as 120
 - Clear “Enable SSH access” (off)

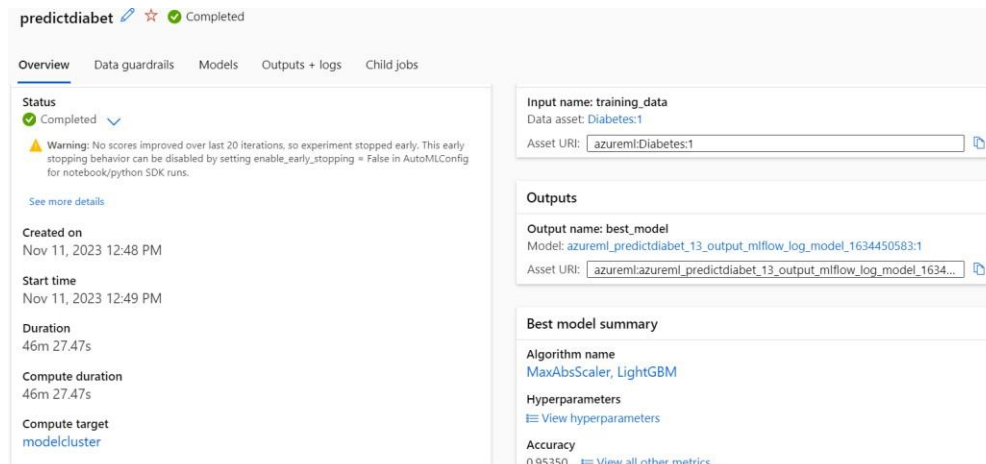
- Click Create. (It will take some time to be created.)
- Click on **Submit training job** button. Wait for the run status to change to completed.



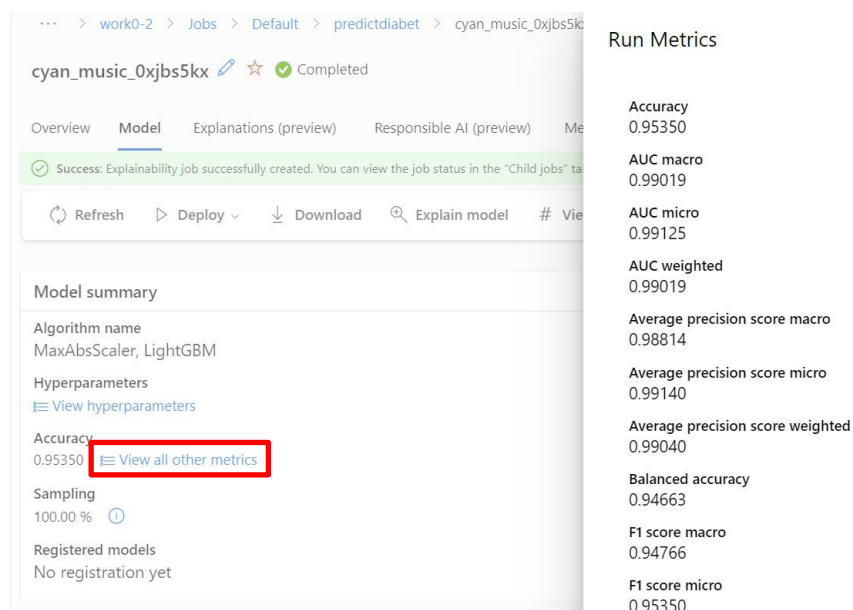
2. When status is *Running*, view the **Models+child job** tab and observe the execution of every potential combination of training algorithms and pre-processing steps and their performance. The page automatically refreshes periodically, but you can also select **Refresh**. It might take 10 minutes or so before models start to appear, as the cluster nodes must be initialized before training can begin.
3. Wait for the job to finish (completed). It might take a while.

Review the best model

1. When the status changes to completed, you can see the best model summary and algorithm in the **Model** tab.



2. Next to the *Accuracy* value, select **View all other metrics** to see values of other possible evaluation metrics.

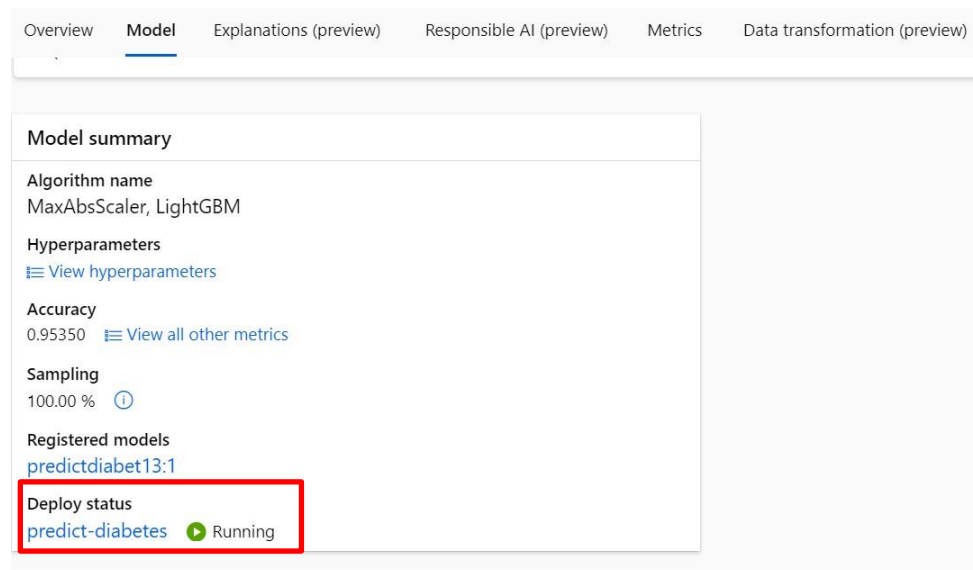


3. Select the **Metrics** tab, click on >> and select the **accuracy, accuracy table, confusion matrix, f1_score_weighted**.
4. Review the charts which show the performance of the model.

Deploy a predictive service

1. On the **Model** tab, select the **Deploy** button and use the **Deploy to web service** option to deploy the model with the following settings:
 - **Name:** predict-diabetes
 - **Description:** Predict diabetes with automl
 - **Compute type:** Azure Container Instance
 - **Enable authentication:** Selected
2. Click on **Deploy** button. Wait for the deployment to start (this may take a few seconds.)

Then, in the **Model summary** section, observe the **Deploy status** for the **predict-diabetes** service, which should be **Running**. Wait for this status to change to **Succeeded**, which may take some time. You may need to select **Refresh** periodically.



Test the deployed service

Now you can test your deployed service.

1. On the left-hand menu, select **Endpoints** (Under Assets section)
2. Then click on **predict-diabetes**.
3. When the **predict-diabetes** endpoint opens, view the **Test** tab.
4. In the **Input data to test real-time endpoint** pane, enter the following values for the fields:

The screenshot shows the 'predict-diabetes' endpoint interface with the 'Test' tab selected. The 'data' dropdown is open, showing a list of input fields with their corresponding values entered:

Field	Value
PatientID	130000
Pregnancies	1
PlasmaGlucose	90
DiastolicBloodPressure	95
TricepsThickness	10
SerumInsulin	250
BMI	25.5
DiabetesPedigree	0.1
Age	25

Click on the **Test** button on the top of the page.

5. The test pane took the input data and used the model you trained to predict the status of patient. 1 means Diabetic and 0 means non-diabetic.

Input data to test endpoint

Test

Test result

Select editor type

☒ Form editor ☐ JSON editor

data

PatientID

130000

Pregnancies

1

PlasmaGlucose

90

```
{ 1 item
  "Results": [ 1 item
    0 : int 0
  ]
}
```

You have just tested a service that is ready to be connected to a client application using the credentials in the **Consume** tab. We will end the lab here. You are welcome to continue to experiment with the service you just deployed.

Clean-up

The web service you created is hosted in an *Azure Container Instance*. If you don't intend to experiment with it further, you should delete the endpoint to avoid accruing unnecessary Azure usage. You should also stop the Compute instance until you need it again.

1. In [Azure Machine Learning studio](#), on the **Endpoints** tab, select the **predict-diabetes** endpoint. Then select **Delete** and confirm that you want to delete the endpoint.
2. On the **Compute** page, on the **Compute Instances** tab, select your compute instance and then select **Stop**.

[!NOTE] Stopping your compute ensures your subscription won't be charged for compute resources. You will however be charged a small amount for data storage as long as the Azure Machine Learning workspace exists in your subscription. If you have finished exploring Azure Machine Learning, you can delete the Azure Machine Learning workspace and associated resources. However, if you plan to complete any other labs in this series, you will need to recreate it.

To delete your workspace:

1. In the [Azure portal](#), click on the **Resource groups**, open the resource group you specified when creating your Azure Machine Learning workspace by clicking on its name.
2. Click **Delete resource group**, type the resource group name to confirm you want to delete it, and select **Delete**.